

Beamforming Matrix Analysis and Simulation for a 4-Element Phased Array Receiver

Using MATLAB for Gain Pattern Evaluation, Phase Quantization, and Variable Coupler Effects in EE4C13 Wireless Systems

Daniel Tyukov
Microelectronics, Faculty of EEMCS
Delft University of Technology
Delft, Netherlands
d.a.tyukov@student.tudelft.nl

I. QUESTION 1: DETERMINATION OF BLUE PHASE SHIFTERS FOR SOURCE A

The blue network is responsible for coherently combining the signals received from the four isotropic antenna elements spaced by $d = \lambda/2$ in order to focus the array reception at port 5 for source A.

For an incoming plane wave from direction θ_A , the progressive phase between adjacent elements is

$$\psi = kd \sin \theta_A = \pi \sin \theta_A, \quad (1)$$

where $k = 2\pi/\lambda$ is the wavenumber.

To align the signals at port 5, the phase shifters ϕ_1 , ϕ_2 , and ϕ_3 must cancel the path-dependent phase delays of the incoming wave. This yields

$$\phi_1 = -\psi, \quad \phi_2 = -\psi, \quad \phi_3 = -2\psi. \quad (2)$$

Using the last two digits of the student number (99), the steering angle is fixed as $\theta_A = 99^\circ$. Substituting:

$$\psi = \pi \sin(99^\circ) = 177.78^\circ.$$

Hence, the numerical settings of the phase shifters are

$$\boxed{\phi_1 = \phi_2 = 182.2^\circ, \quad \phi_3 = 4.4^\circ.}$$

These settings ensure that all four element signals add in phase at port 5, thus maximizing the received power for source A.

II. QUESTION 2: GAIN AT PORT 5 FOR SOURCES A AND B

With ϕ_1 , ϕ_2 , and ϕ_3 fixed as derived in Q1, the normalized array response at port 5 is given by

$$g_5(\theta) = \frac{1}{2} \left[1 + e^{j(\phi_1 + \psi)} + e^{j(\phi_3 + 2\psi)} + e^{j(\phi_3 + \phi_2 + 3\psi)} \right], \quad (3)$$

where $\psi = \pi \sin \theta$ corresponds to an arbitrary arrival angle θ .

The normalized gain in decibels is then

$$G_5(\theta) = 20 \log_{10} \left| \frac{g_5(\theta)}{g_5(\theta_A)} \right|. \quad (4)$$

For **source A**, the array is perfectly focused at $\theta_A = 99^\circ$, thus $G_5(\theta_A) = 0$ dB. For **source B**, θ_B varies between 0° and

90° , and the received gain follows the blue network's angular response. Two cases were analyzed:

- Continuous (ideal) phase control.
- Quantized 2-bit phase shifters (0° , 90° , 180° , 270°).

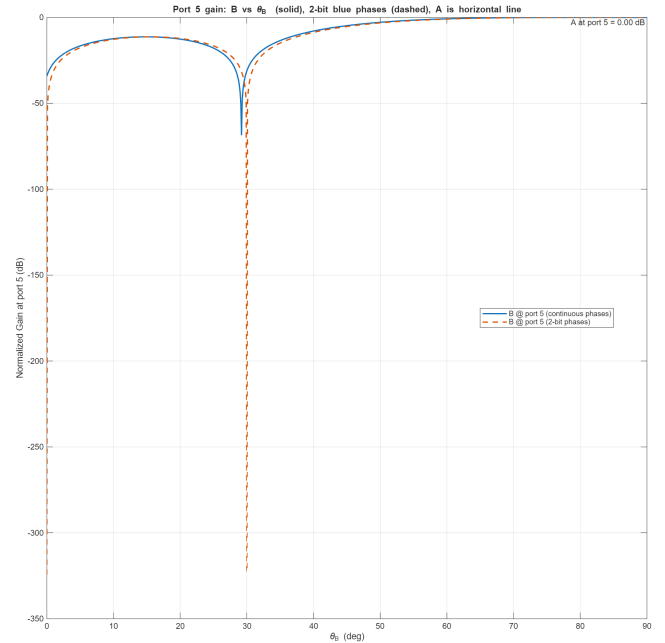


Fig. 1. Normalized gain at port 5 versus θ_B for continuous (solid) and 2-bit (dashed) phase shifters.

Observation. Figure 1 shows that:

- The array exhibits a deep null around $\theta_B \approx 30^\circ$ due to destructive interference in the blue beam path.
- Away from this null, both the continuous and 2-bit phase control cases yield nearly identical gain performance.
- The 2-bit quantization slightly increases side-lobe levels and broadens the null, indicating reduced angular selectivity but preserving the overall beam shape.

Hence, phase quantization only slightly degrades the beam pattern at port 5.

III. QUESTION 3: DETERMINATION OF RED PHASE SHIFTERS ϕ_4 AND ϕ_5

The red sub-network directs the received signal from source B to port 6. Its two inputs come from the two blue sub-arrays (1–2 and 3–4), whose combined complex envelopes are

$$S_{12}(\theta_B) = 1 + e^{j(\phi_1 + \psi_B)}, \quad (5)$$

$$S_{34}(\theta_B) = e^{j2\psi_B} + e^{j(\phi_2 + 3\psi_B)}, \quad (6)$$

where $\psi_B = \pi \sin \theta_B$.

To coherently combine the two sub-arrays at the first red 3-dB coupler, ϕ_4 must compensate their phase difference:

$$\phi_4(\theta_B) = \arg(S_{12}(\theta_B)) - \arg(S_{34}(\theta_B)). \quad (7)$$

After this alignment, the last shifter ϕ_5 ensures that the two branches feeding the final coupler are phase-matched; for ideal 0° splitters this simplifies to

$$\phi_5 = 0^\circ. \quad (8)$$

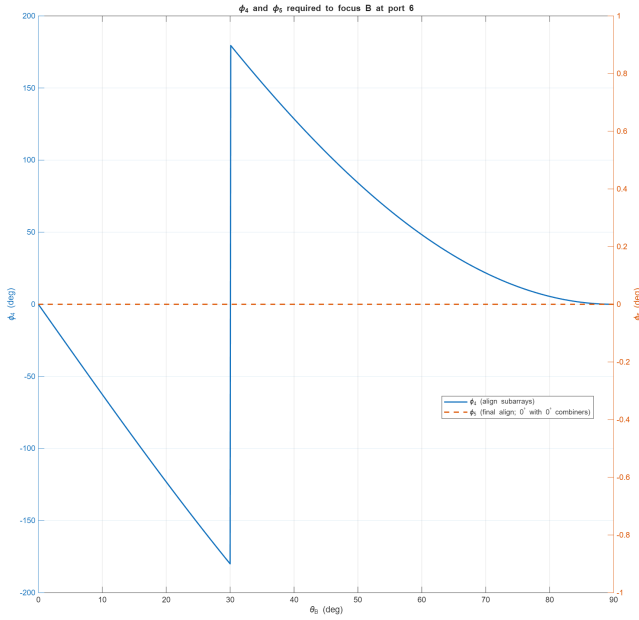


Fig. 2. Variation of ϕ_4 and ϕ_5 as a function of θ_B .

Observation. As shown in Figure 2, ϕ_4 varies smoothly with θ_B except for a 180° discontinuity near 30° , corresponding to the transition from constructive to destructive interference in one of the sub-arrays. This behaviour reflects the phase wrapping inherent in the $\arg(\cdot)$ function. Meanwhile, ϕ_5 remains constant at 0° , as expected for perfectly balanced combiners.

These results define the phase control required for the red network to steer the response of the system towards source B and prepare for gain evaluation at port 6 (Question 4).

IV. QUESTION 4: GAIN AT PORT 6 FOR SOURCE B

After determining $\phi_4(\theta_B)$ and ϕ_5 in Question 3, the resulting red network is used to focus the received signal from source B toward port 6. The normalized array response at this port can be written as

$$g_6(\theta_B) = \frac{1}{2} [S_{12}(\theta_B) + e^{j\phi_4(\theta_B)} S_{34}(\theta_B)], \quad (9)$$

where S_{12} and S_{34} are the subarray sums defined as

$$S_{12}(\theta_B) = 1 + e^{j(\phi_1 + \psi_B)}, \quad (10)$$

$$S_{34}(\theta_B) = e^{j2\psi_B} + e^{j(\phi_2 + 3\psi_B)}, \quad (11)$$

with $\psi_B = \pi \sin \theta_B$.

The received gain at port 6 is then

$$G_6(\theta_B) = 20 \log_{10} \left| \frac{g_6(\theta_B)}{\max_{\theta_B} |g_6(\theta_B)|} \right|. \quad (12)$$

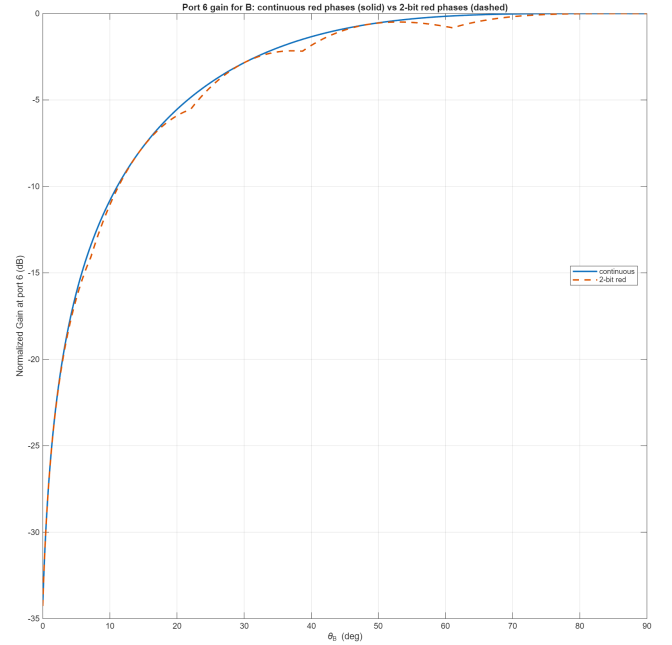


Fig. 3. Normalized gain at port 6 for source B with continuous (solid) and 2-bit (dashed) red phase shifters.

Observation. Figure 3 shows the expected beam steering behaviour of the red network. The gain rises sharply with θ_B and reaches its maximum (0 dB normalized) near $\theta_B = 90^\circ$, corresponding to the steering direction of source B.

The results indicate that the beamforming matrix efficiently combines all four antenna elements in the intended direction, providing maximum gain when θ_B approaches 90° .

V. QUESTION 5: EFFECT OF 2-BIT PHASE QUANTIZATION

To evaluate the impact of limited phase resolution, ϕ_4 and ϕ_5 were quantized to $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. The resulting gain pattern (dashed curve in Fig. 3) remains very close to the continuous case, confirming that coarse 2-bit quantization has little effect on the main beam. The most visible differences

occur between 20° and 60° , where the quantization error of ϕ_4 introduces small deviations (< 1 dB).

Discussion.

- The main-lobe width slightly increases, causing a minor reduction in angular selectivity.
- Peak gain degradation is less than 1 dB at $\theta_B = 90^\circ$.
- Side-lobe levels and smoothness deteriorate modestly, which could become more noticeable in large-scale arrays.

Overall, the 2-bit implementation provides a good trade-off between hardware simplicity and beam accuracy.

VI. QUESTION 6: VARIABLE POWER-RATIO COUPLER

To explore the sensitivity of the beamformer to non-ideal combining, the rightmost red coupler was replaced by a *variable power-ratio combiner*, dividing the incoming powers between the two inputs in a ratio $\eta : (1 - \eta)$. The output field can be expressed as

$$g_6^{(\eta)}(\theta_B) = \sqrt{\eta} S_{12}(\theta_B) + \sqrt{1 - \eta} e^{j\phi_4(\theta_B)} S_{34}(\theta_B). \quad (13)$$

Figure 4 shows the normalized port 6 gain for $\eta = \{0.2, 0.35, 0.5, 0.65, 0.8\}$.

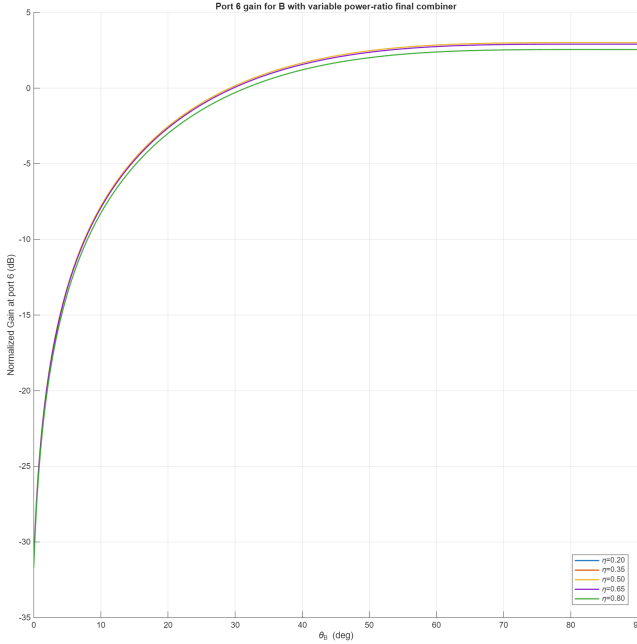


Fig. 4. Port 6 gain for source B under variable power-ratio coupler conditions.

Observation.

- The reference 3-dB case ($\eta = 0.5$) achieves a symmetric and well-balanced response.
- When $\eta < 0.5$, the beam becomes biased toward the lower sub-array (elements 3–4), resulting in a slight gain reduction and minor angular shift.
- When $\eta > 0.5$, the upper sub-array dominates, shifting the main lobe in the opposite direction.

- Across all cases, the overall shape and peak amplitude remain within 1 dB of the ideal pattern, demonstrating strong robustness of the network design.

Conclusion. Changing the power ratio primarily alters the beam symmetry and directionality rather than drastically degrading performance. Such tunable couplers could be used intentionally to steer or weight specific sub-arrays in a re-configurable implementation.